Abstract Algebra Homework1

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Problem 1

Part (a)

Claim. *The number of functions* $\{1, \ldots, n\} \to \{1, \ldots, m\}$ *is* m^n.

Proof. We proceed by induction on n.

Base case ($n = 0$): $\{1, \ldots, 0\} = \emptyset$, and there is exactly one function from the empty set to any set (the empty function). Indeed $m^0 = 1$.

Inductive step: Assume every function $\{1, \ldots, n\} \to \{1, \ldots, m\}$ is counted by m^n. A function $f : \{1, \ldots, n+1\} \to \{1, \ldots, m\}$ is determined by its restriction $f|_{\{1,\ldots,n\}}$ together with the value $f(n+1)$. By the inductive hypothesis there are m^n choices for the restriction, and m choices for $f(n+1)$. These choices are independent, so by the multiplication principle there are $m^n \cdot m = m^{n+1}$ functions in total. □

Part (b)

Claim. *For* $n \leq m$*, the number of injective functions* $\{1, \ldots, n\} \to \{1, \ldots, m\}$ *is* $\dfrac{m!}{(m-n)!}$.

Proof. We proceed by induction on n.

Base case ($n = 0$): There is one injective function from $\emptyset$ (the empty function), and $\frac{m!}{m!} = 1$.

Inductive step: Suppose the count for n is $\frac{m!}{(m-n)!}$, and let $n + 1 \leq m$. An injective function $f : \{1, \ldots, n+1\} \to \{1, \ldots, m\}$ is determined by an injective function $f|_{\{1,\ldots,n\}} : \{1, \ldots, n\} \to \{1, \ldots, m\}$ together with a value $f(n+1) \in \{1, \ldots, m\} \setminus \mathrm{Im}(f|_{\{1,\ldots,n\}})$. Since $f|_{\{1,\ldots,n\}}$ is injective, its image has size n, leaving $m - n$ valid choices for $f(n+1)$. By the inductive hypothesis the count is

$$\frac{m!}{(m-n)!} \cdot (m-n) = \frac{m!}{(m-(n+1))!}.$$

□

Part (c)

Claim. *Every injective function* $f : \{1, \ldots, n\} \to \{1, \ldots, n\}$ *is bijective. Consequently the number of bijections* $\{1, \ldots, n\} \to \{1, \ldots, n\}$ *is* $n!$.

Proof. Since f is injective, $|\operatorname{Im}(f)| = n$. But $\operatorname{Im}(f) \subseteq \{1, \ldots, n\}$ has at most n elements, so $\operatorname{Im}(f) = \{1, \ldots, n\}$, i.e. f is surjective. Hence f is bijective.

For the count: a bijection $\{1, \ldots, n\} \to \{1, \ldots, n\}$ is in particular an injective function from $\{1, \ldots, n\}$ to $\{1, \ldots, n\}$, and by the above every such injection is a bijection. By part (b) with $m = n$, the number of injections is $\frac{n!}{(n-n)!} = n!$. $\square$

Problem 2

Part (a)

Claim. *The number of subsets of* $\{1, \ldots, n\}$ *is* 2^n.

Proof. We proceed by induction on n.

Base case ($n = 0$): $\mathcal{P}(\emptyset) = \{\emptyset\}$ has $1 = 2^0$ element.

Inductive step: Every subset $S \subseteq \{1, \ldots, n+1\}$ either contains $n+1$ or does not. If it does not, S is a subset of $\{1, \ldots, n\}$; if it does, $S = T \cup \{n+1\}$ for a unique subset $T \subseteq \{1, \ldots, n\}$. By the inductive hypothesis there are 2^n subsets of $\{1, \ldots, n\}$ in each case, giving $2 \cdot 2^n = 2^{n+1}$ subsets of $\{1, \ldots, n+1\}$ in total. □

Part (b)

Claim. *The number of subsets of* $\{1, \ldots, n\}$ *of size* k *is* $\binom{n}{k}$.

Proof. We count the number of injective functions $g : \{1, \ldots, k\} \to \{1, \ldots, n\}$ in two ways.

By Problem 1(b), this number is $\frac{n!}{(n-k)!}$.

Alternatively, each such injection is an *ordered* selection of k distinct elements of $\{1, \ldots, n\}$. Every k-element subset S gives rise to exactly $k!$ such injections (one per ordering of S, by Problem 1(c)). If N is the number of k-element subsets, then

$$N \cdot k! = \frac{n!}{(n-k)!},$$

so $N = \dfrac{n!}{k!(n-k)!} = \dbinom{n}{k}$. □

Part (c)

Claim. $2^n = \displaystyle\sum_{k=0}^{n} \binom{n}{k}$.

Proof. By part (a), $\{1, \ldots, n\}$ has 2^n subsets in total. Every subset has size k for exactly one $k \in \{0, 1, \ldots, n\}$. By part (b), the number of subsets of size k is $\binom{n}{k}$. Summing over all k gives

$$2^n = \sum_{k=0}^{n} \binom{n}{k}.$$ □

Problem 3

Let $f : X \to Y$ and $g : Y \to Z$.

Part (a)

Claim. *If f and g are injective, then $g \circ f$ is injective. If f and g are surjective, then $g \circ f$ is surjective.*

Proof. **Injectivity.** Suppose f and g are injective. Let $x_1, x_2 \in X$ with $(g \circ f)(x_1) = (g \circ f)(x_2)$, i.e. $g(f(x_1)) = g(f(x_2))$. Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$.

Surjectivity. Suppose f and g are surjective. Let $z \in Z$. Since g is surjective, there exists $y \in Y$ with $g(y) = z$. Since f is surjective, there exists $x \in X$ with $f(x) = y$. Then $(g \circ f)(x) = g(y) = z$. □

Part (b)

Claim. *If $g \circ f$ is injective, then f is injective. If $g \circ f$ is surjective, then g is surjective.*

Proof. **Injectivity.** Suppose $g \circ f$ is injective. Let $x_1, x_2 \in X$ with $f(x_1) = f(x_2)$. Then $g(f(x_1)) = g(f(x_2))$, i.e. $(g \circ f)(x_1) = (g \circ f)(x_2)$. Since $g \circ f$ is injective, $x_1 = x_2$.

Surjectivity. Suppose $g \circ f$ is surjective. Let $z \in Z$. There exists $x \in X$ with $(g \circ f)(x) = z$, i.e. $g(f(x)) = z$. Taking $y = f(x) \in Y$, we have $g(y) = z$. □

Part (c)

Claim. *g need not be injective even if $g \circ f$ is; f need not be surjective even if $g \circ f$ is.*

Proof. Let $X = \{1\}$, $Y = \{1, 2\}$, $Z = \{1\}$, and define $f(1) = 1$, $g(1) = g(2) = 1$. Then $g \circ f : \{1\} \to \{1\}$ is injective (and bijective), but g is not injective since $g(1) = g(2)$.

For the second statement, keep the same example: $g \circ f$ is surjective (it hits $1 \in Z$), but $f : \{1\} \to \{1, 2\}$ is not surjective since $2 \notin \mathrm{Im}(f)$. □

Problem 4

Let $f : X \to Y$ be a function. A *right inverse* of f is $g : Y \to X$ with $f \circ g = \mathrm{id}_Y$.

Part (a)

Claim. *If f has a right inverse, then f is surjective.*

Proof. Let $g : Y \to X$ with $f \circ g = \mathrm{id}_Y$. For any $y \in Y$, set $x = g(y) \in X$; then $f(x) = f(g(y)) = (f \circ g)(y) = \mathrm{id}_Y(y) = y$. So every y is in the image of f. □

Part (b)

Claim. *If f is surjective, then f has a right inverse.*

Proof. Since f is surjective, for every $y \in Y$ the preimage $f^{-1}(y) = \{x \in X : f(x) = y\}$ is non-empty. Using the axiom of choice, pick one element $g(y) \in f^{-1}(y)$ for each $y \in Y$. This defines a function $g : Y \to X$ satisfying $f(g(y)) = y$ for all y, i.e. $f \circ g = \mathrm{id}_Y$. □

Part (c)

Claim. *A surjective function need not have a unique right inverse.*

Proof. Let $X = \{1, 2\}$, $Y = \{1\}$, and $f(1) = f(2) = 1$. Then f is surjective. Any function $g : Y \to X$ satisfies $f(g(1)) = 1$, so both $g_1(1) = 1$ and $g_2(1) = 2$ are right inverses. These are distinct, so the right inverse is not unique. □

Problem 5

Claim. *Let $\sim$ be an equivalence relation on X and let $x, y \in X$. Then either $[x] \cap [y] = \emptyset$ or $[x] = [y]$.*

Proof. Suppose $[x] \cap [y] \neq \emptyset$, and let $z \in [x] \cap [y]$. Then $x \sim z$ and $y \sim z$.

We show $[x] \subseteq [y]$. Let $w \in [x]$, so $x \sim w$. By symmetry $z \sim x$; by transitivity $z \sim w$. Since $y \sim z$ and $z \sim w$, transitivity gives $y \sim w$, i.e. $w \in [y]$.

By symmetry of the argument (swapping x and y), $[y] \subseteq [x]$. Therefore $[x] = [y]$. $\square$

Problem 6

Claim. *The function* $\Gamma : \mathrm{EQ}(X) \to \mathrm{PT}(X)$ *defined by* $\Gamma(E) = \{[x]_E : x \in X\}$ *is a bijection.*

Proof. We construct a two-sided inverse $\Delta : \mathrm{PT}(X) \to \mathrm{EQ}(X)$.

Given a partition $\mathcal{C}$ of X, define the relation $\Delta(\mathcal{C})$ on X by

$$x \,\Delta(\mathcal{C})\, y \iff \exists\, C \in \mathcal{C},\ x, y \in C.$$

$\Delta(\mathcal{C})$ is an equivalence relation.

- *Reflexivity*: Since $\mathcal{C}$ covers X, every x belongs to some $C \in \mathcal{C}$, so $x \,\Delta(\mathcal{C})\, x$.
- *Symmetry*: If $x, y \in C$, then $y, x \in C$.
- *Transitivity*: If $x, y \in C$ and $y, z \in C'$ for $C, C' \in \mathcal{C}$, then $y \in C \cap C'$. Since $\mathcal{C}$ is a partition, $C = C'$, so $x, z \in C$, giving $x \,\Delta(\mathcal{C})\, z$.

$\Delta \circ \Gamma = \mathrm{id}_{\mathrm{EQ}(X)}$**.** Let $E \in \mathrm{EQ}(X)$. We have $\Gamma(E) = \{[x]_E : x \in X\}$, and $\Delta(\Gamma(E))$ is the relation "x and y lie in the same equivalence class of E", which is exactly E.

$\Gamma \circ \Delta = \mathrm{id}_{\mathrm{PT}(X)}$**.** Let $\mathcal{C}$ be a partition. The equivalence classes of $\Delta(\mathcal{C})$ are exactly the parts $C \in \mathcal{C}$: indeed, $[x]_{\Delta(\mathcal{C})} = \{y \in X : x \sim y\}$ is the unique block $C \in \mathcal{C}$ containing x. So $\Gamma(\Delta(\mathcal{C})) = \mathcal{C}$.

Since Δ is a two-sided inverse of Γ, the function Γ is a bijection. □

Problem 7

The relation on $\mathbb{N}^2$: $(a,b) \sim (c,d) \iff a+d = b+c$.

Part (a)

Claim. *$\sim$ is an equivalence relation on $\mathbb{N}^2$.*

Proof. *Reflexivity*: $a+b = b+a$, so $(a,b) \sim (a,b)$.
Symmetry: If $a+d = b+c$, then $c+b = d+a$, so $(c,d) \sim (a,b)$.
Transitivity: Suppose $(a,b) \sim (c,d)$ and $(c,d) \sim (e,f)$, i.e. $a+d = b+c$ and $c+f = d+e$. Adding: $a+d+c+f = b+c+d+e$. Cancel $c+d$: $a+f = b+e$, so $(a,b) \sim (e,f)$. □

Part (b)

Claim. *If $a = b+k$ for some $k \in \mathbb{N}$, then $(a,b) \sim (k,0)$; if $b = a+k$, then $(a,b) \sim (0,k)$.*

Proof. If $a = b+k$: check $a+0 = b+k$, i.e. $a+0 = b+k$.
If $b = a+k$: check $a+k = b+0$, i.e. $a+k = b$.
Every pair $(a,b) \in \mathbb{N}^2$ satisfies exactly one of $a > b$, $a = b$, or $a < b$:

- $a > b$: $k = a-b > 0$, so $(a,b) \sim (k,0)$.
- $a = b$: $k = 0$, so $(a,b) \sim (0,0) = (k,0)$.
- $a < b$: $k = b-a > 0$, so $(a,b) \sim (0,k)$.

Thus $\{(k,0) : k \in \mathbb{N}\} \cup \{(0,k) : k \in \mathbb{N}, k \geq 1\}$ is a complete set of representatives. □

Part (c)

Claim. *The operation p descends to a well-defined binary operation on $\mathbb{Z} = \mathbb{N}^2/\sim$, making $(\mathbb{Z}, +)$ a group.*

Proof. **Well-definedness.** Suppose $(a,b) \sim (a',b')$ and $(c,d) \sim (c',d')$, i.e. $a+b' = b+a'$ and $c+d' = d+c'$. We need $(a+c, b+d) \sim (a'+c', b'+d')$, i.e.

$$(a+c)+(b'+d') = (b+d)+(a'+c'),$$

which holds by adding the two hypothesis equalities.

Group axioms. Write $[a,b]$ for the class of (a,b).

- *Associativity*: Addition in $\mathbb{N}$ is associative, so $[a,b]+([c,d]+[e,f]) = [a+c+e, b+d+f] = ([a,b]+[c,d])+[e,f]$.
- *Identity*: $[0,0]$ is the identity: $[a,b]+[0,0] = [a,b]$.
- *Inverses*: $[a,b]+[b,a] = [a+b, b+a] \sim [0,0]$ since $(a+b)+0 = (b+a)+0$. So $[b,a]$ is the inverse of $[a,b]$.

□

Part (d)

Claim. *The operation m descends to a well-defined binary operation on $\mathbb{Z}$, making $(\mathbb{Z}, \cdot)$ a monoid but not a group.*

Proof. **Well-definedness.** Suppose $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$, so $a+b' = b+a'$ and $c + d' = d + c'$. We need

$$(ac + bd) + (a'd' + b'c') = (ad + bc) + (a'c' + b'd').$$

Expanding both sides and using the two hypotheses (with some algebraic manipulation using commutativity and distributivity in $\mathbb{N}$), this identity holds. (The key identity is $(a - b)(c - d) = (a' - b')(c' - d')$ over integers, which follows since $a - b = a' - b'$ and $c - d = c' - d'$ in $\mathbb{Z}$.)

Monoid axioms.

- *Associativity*: Follows from associativity and distributivity in $\mathbb{N}$.
- *Identity*: $[1, 0]$ acts as identity: $[a, b] \cdot [1, 0] = [a \cdot 1 + b \cdot 0,\ a \cdot 0 + b \cdot 1] = [a, b]$.

Not a group. The class $[0, 0]$ (the additive identity, representing 0) satisfies $[0, 0] \cdot [c, d] = [0, 0]$ for all $[c, d]$, so it cannot have a multiplicative inverse (as $[0, 0] \neq [1, 0]$). Hence $(\mathbb{Z}, \cdot)$ is not a group. □

Problem 8

Claim. *Let $(G, *)$ be a group and $a, b \in G$. There exist unique $x, y \in G$ with $a * x = b$ and $y * a = b$.*

Proof. **Existence.** Set $x = a^{-1} * b$. Then $a * x = a * (a^{-1} * b) = (a * a^{-1}) * b = e * b = b$. Similarly, $y = b * a^{-1}$ satisfies $y * a = (b * a^{-1}) * a = b * (a^{-1} * a) = b * e = b$.

Uniqueness for x. If $a * x = b$ and $a * x' = b$, then $a * x = a * x'$. Left-multiplying by a^{-1}: $x = x'$.

Uniqueness for y. If $y * a = b$ and $y' * a = b$, then $y * a = y' * a$. Right-multiplying by a^{-1}: $y = y'$. □

Problem 9

Claim. *Let $(G, *)$ be a group and $a \in G$. Then $(a^{-1})^{-1} = a$.*

Proof. By definition of inverse, a^{-1} is the unique element satisfying $a^{-1} * a = e$ and $a * a^{-1} = e$. Rewriting:

$$a^{-1} * a = e \quad \text{and} \quad a * a^{-1} = e.$$

These two equations say exactly that a is an inverse of a^{-1}. Since inverses in a group are unique (which follows from Problem 8 applied with $b = e$), we conclude $(a^{-1})^{-1} = a$. $\square$

Problem 10

Part (a)

Claim. *The multiplication* $[a]_p * [b]_p := [ab]_p$ *on* $\mathbb{Z}/p\mathbb{Z}$ *is well-defined.*

Proof. Suppose $[a]_p = [a']_p$ and $[b]_p = [b']_p$, i.e. $p \mid (a - a')$ and $p \mid (b - b')$. Then

$$ab - a'b' = ab - a'b + a'b - a'b' = b(a - a') + a'(b - b').$$

Since p divides each term, $p \mid (ab - a'b')$, i.e. $[ab]_p = [a'b']_p$. □

Part (b)

Claim. *The following are equivalent:*

(i) p *is prime.*

(ii) $(\mathbb{Z}/p\mathbb{Z} \setminus \{[0]_p\}, *)$ *is a group.*

Proof. (**i**) ⇒ (**ii**). Assume p is prime.

Closure: If $p \nmid a$ and $p \nmid b$, then since p is prime, $p \nmid ab$, so $[ab]_p \neq [0]_p$.

Associativity: Inherited from $\mathbb{Z}$.

Identity: $[1]_p$ is the identity, and $[1]_p \neq [0]_p$ since $p > 1$.

Inverses: Let $[a]_p \neq [0]_p$, so $\gcd(a, p) = 1$ (since p is prime). By Bézout's theorem there exist $s, t \in \mathbb{Z}$ with $sa + tp = 1$, giving $[s]_p * [a]_p = [sa]_p = [1 - tp]_p = [1]_p$. So $[s]_p$ is the inverse of $[a]_p$.

(**ii**) ⇒ (**i**). We prove the contrapositive: if p is not prime (and $p > 1$), then $\mathbb{Z}/p\mathbb{Z} \setminus \{[0]_p\}$ is not closed under $*$.

Since p is not prime, write $p = ab$ with $1 < a, b < p$. Then $[a]_p \neq [0]_p$ and $[b]_p \neq [0]_p$, but $[a]_p * [b]_p = [ab]_p = [p]_p = [0]_p$. So the set is not closed under multiplication, and in particular cannot be a group.

(If $p = 1$, then $\mathbb{Z}/p\mathbb{Z} = \{[0]_p\}$ and $\mathbb{Z}/p\mathbb{Z} \setminus \{[0]_p\} = \emptyset$ has no identity, so it is not a group either.) □